

Tim J. Schoonbeek

PHD CANDIDATE · COMPUTER VISION · MULTIMODAL MACHINE LEARNING

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Summary

PhD candidate and Applied ML Researcher at ASML and TU/e (graduating November 25). I build scalable computer vision systems in industrial settings for augmented reality applications, ranging from multimodal video models for step recognition to assembly object detection and error localization. Proven track record translating bleeding-edge AI into real-world prototypes, as demonstrated through patents and publications in computer vision (WACV), robotics (ICRA), and autonomous driving (IV). I am driven to build production-grade ML systems that scale, deliver measurable impact, and bridge the gap between research breakthroughs and actual products.

TECHNICAL SKILLS

Deep learning: CV, 3D perception, ML, augmented reality, HCI, video understanding, transformers, convnets, contrastive learning.
Languages & tools: Python, PyTorch, NumPy, OpenCV, Unity, Pandas, Azure, Git, (C++).

Experience

Microsoft Research

Redmond, USA

RESEARCH INTERN: INTERACTIVE MULTIMODAL AI SYSTEMS

June 2025 - Sept. 2025

- Developing real-time multimodal systems with LLM-powered perception to create benchmarks for physically situated intelligence.
- Building models of social cues (turn-taking, engagement, attention) from large datasets while collaborating with cross-disciplinary teams.

ASML Research

Eindhoven, the Netherlands

VISITING RESEARCHER: PHD ON ADVANCED AUGMENTED REALITY SOLUTIONS FOR AI-BASED SERVICITIZATION

Sept. 2021 - Nov. 2025

- PhD project on automatic extraction of relevant and meaningful information from procedural actions within an industrial setting.
- Filed patents, published peer-reviewed articles, and contributed to the ASML Technology Conference, the world's largest developer event.
- Built egocentric industrial dataset & evals (IndustReal); released tasks and metrics; benchmarked with several backbones (5k+ downloads).

Honda Research Institute Europe GmbH

Frankfurt, Germany

MASTER INTERNSHIP: INTERACTION-AWARE TRAJECTORY PREDICTION USING GRAPH NEURAL NETWORKS

Mar. 2020 - July 2020

- Researched spatio-temporal graph neural networks for trajectory prediction of surrounding vehicles in highway driving scenarios.

Automotive Technology InMotion (Student Team)

Helmond, the Netherlands

RACE ENGINEER - PART-TIME

Sept. 2018 - Sept. 2019

- Performed and optimized drive cycle simulations for an electric endurance racing car.

Education

Eindhoven University of Technology

Eindhoven, Netherlands

PHD ON ADVANCED AUGMENTED REALITY SOLUTIONS FOR AI-BASED SERVICITIZATION (ELECTRICAL ENGINEERING FACULTY)

Sept. 2021 - Nov. 2025

- Doctoral project funded by ASML on industrial procedure understanding using computer vision and machine learning algorithms.
- Project on the intersection of computer vision, human-computer interaction, and automation.
- Focus on scalable, multimodal computer-vision systems for procedure understanding and creating relevant industrial benchmarks.

University of Cambridge

Cambridge, United Kingdom

VISITING RESEARCHER

Oct. 2024 - April 2025

- Conducted research on enhancing human-robot interaction, focusing on AI systems to assist industrial operators in procedure execution.
- Designed perception algorithms for error localization for a real-time deployment on AR glasses; conducted user studies (N=24).
- Reached the finals (top 2.5%) in the OxBridge AI Startup Challenge, a competition between students from Cambridge and Oxford.

International Summer School

Matera, Italy

EXTENDED REALITY AND ARTIFICIAL INTELLIGENCE

July 2023

- Pro-actively took up the role as a link between the programmers and designers in a multidisciplinary project, besides my technical roles.
- Won the *best project award* for our work on an XR app for engagement with cultural heritage.

Eindhoven University of Technology

Eindhoven, Netherlands

M.Sc. IN AUTOMOTIVE ENGINEERING (SPECIALIZATION: MOBILE PERCEPTION SYSTEMS)

Feb. 2019 - May 2021

- Earned the *cum laude* honours distinction for a 4.0 GPA (8.5/10).
- Master thesis "Learning to Predict Collision Risk from Simulated Optical Flow" awarded a 9.0/10.0. Conducted research on sim-to-real domain generalization by design, rather than through explicit domain adaptation techniques.
- Published paper, singled out among the top 5% of submissions for *oral presentation* at the 2022 IEEE Intelligent Vehicles Symposium.

Eindhoven University of Technology

Eindhoven, Netherlands

B.Sc. IN ELECTRICAL ENGINEERING (AUTOMOTIVE TRACK)

Sept. 2015 - Feb. 2019

- Bachelor thesis on depth estimation from disparity and segmented images awarded a 8.5/10.

Highlighted publications

WACV 2024	Tim J. Schoonbeek <i>et al.</i> , IndustReal: A Dataset for Procedure Step Recognition Handling Execution Errors in Egocentric Videos in an Industrial-Like Setting
CVIU (J.)	Tim J. Schoonbeek <i>et al.</i> , Learning to Recognize Correctly Completed Procedure Steps in Egocentric Assembly Videos through Spatio-Temporal Modeling
ICRA & RA-L	Tim J. Schoonbeek <i>et al.</i> , Supervised Representation Learning towards Generalizable Assembly State Recognition
ECCV-W 2024	Tim J. Schoonbeek and Dan Lehman <i>et al.</i> , Find the Assembly Mistakes: Error Segmentation for Industrial Applications
IEEE IV 2022	Tim J. Schoonbeek <i>et al.</i> , Learning to Predict Collision Risk from Simulated Video Data

Industry Contributions

EP Patent (<i>Publ.</i>)	Tim J. Schoonbeek <i>et al.</i> , Defect Identification and Segmentation without Labeled Data and Image Quality Enhancement
WO Patent (<i>Publ.</i>)	Tim J. Schoonbeek <i>et al.</i> , Contrastive Deep Learning for Scanning Electron Microscope Defect Inspection
ASML TC 2024	Tim J. Schoonbeek <i>et al.</i> , Verifying Procedure States of Assemblies using Contrastive Learning and Synthetic Data
ASML TC 2024	Thomas Van de Moosdijk <i>et al.</i> , Superimposed Work Instructions using Augmented Reality
ASML TC 2023	Tim J. Schoonbeek <i>et al.</i> , Automated Recognition of Procedural Actions for Real-Time Understanding of Service Tasks